

$$\rightarrow -\frac{1}{16\pi^2} \left( -\frac{3}{128} \frac{1}{m_\Sigma^2} S_4 Y^2 (g_1^2 + g_2^2)^2 + \frac{1}{6} g_2^4 LF_{3,0}[m_2] + \frac{1}{4} g_2^4 LF_{4,-1}[m_2] - \frac{4}{45} g_2^4 LF_{5,-2}[m_2] + \right. \\ \left. -\frac{1}{216} \sum_{\mathbf{p}} g_1^4 (4 + 9 c_{2Y}^2) LF_{3,0}[m_d^{\mathbf{p}}] - \frac{1}{144} \sum_{\mathbf{p}} g_1^4 (5 + 6 c_{2Y}^2) LF_{4,-1}[m_d^{\mathbf{p}}] + \right. \\ \left. -\frac{1}{135} \sum_{\mathbf{p}} g_1^4 LF_{5,-2}[m_d^{\mathbf{p}}] - \frac{1}{2} c_{2Y} g_1^2 c_Y^2 Y_d^{\text{pr}} Y_d^{\text{pr}} LF_{3,0}[m_d^{\mathbf{r}}] + \frac{1}{2} c_{2Y} g_1^2 c_Y^2 Y_d^{\text{pr}} Y_d^{\text{pr}} LF_{4,-1}[m_d^{\mathbf{r}}] + \right. \\ \left. -\frac{1}{72} \sum_{\mathbf{p}} g_1^4 (4 + 9 c_{2Y}^2) LF_{3,0}[m_e^{\mathbf{p}}] - \frac{1}{48} \sum_{\mathbf{p}} g_1^4 (5 + 6 c_{2Y}^2) LF_{4,-1}[m_e^{\mathbf{p}}] + \frac{45}{45} \sum_{\mathbf{p}} g_1^4 LF_{5,-2}[m_e^{\mathbf{p}}] - \right. \\ \left. -\frac{1}{2} c_{2Y} g_1^2 c_Y^2 Y_e^{\text{pr}} Y_e^{\text{pr}} LF_{3,0}[m_e^{\mathbf{r}}] + \frac{1}{2} c_{2Y} g_1^2 c_Y^2 Y_e^{\text{pr}} Y_e^{\text{pr}} LF_{4,-1}[m_e^{\mathbf{r}}] + \right. \\ \left. -\frac{1}{144} (36 c_{2Y} c_Y^2 Y_e^{\text{pr}} Y_e^{\text{pr}} (g_1^2 + g_2^2) + \sum_{\mathbf{p}} (g_1^4 (4 + 9 c_{2Y}^2) + 3 g_2^4 (4 - 3 c_{2Y}^2))) LF_{3,0}[m_l^{\mathbf{p}}] + \right. \\ \left. + \left( -\frac{1}{4} c_{2Y} c_Y^2 Y_e^{\text{pr}} Y_e^{\text{pr}} (g_1^2 + g_2^2) - \frac{1}{96} \sum_{\mathbf{p}} (g_1^4 (5 + 6 c_{2Y}^2) + 3 g_2^4 (5 - 2 c_{2Y}^2)) \right) LF_{4,-1}[m_l^{\mathbf{p}}] + \right. \\ \left. + \frac{1}{45} \sum_{\mathbf{p}} (g_1^4 + 3 g_2^4) LF_{5,-2}[m_l^{\mathbf{p}}] + \left( -\frac{1}{4} c_{2Y} c_Y^2 Y_d^{\text{pr}} Y_d^{\text{pr}} (g_1^2 - 3 g_2^2) - \right. \right. \\ \left. \left. -\frac{1}{4} c_{2Y} s_Y^2 Y_u^{\text{pr}} Y_u^{\text{pr}} (g_1^2 + 3 g_2^2) + \frac{1}{432} \sum_{\mathbf{p}} (g_1^4 (4 + 9 c_{2Y}^2) + 27 g_2^4 (4 - 3 c_{2Y}^2)) \right) \right. \\ \left. LF_{3,0}[m_q^{\mathbf{p}}] + \left( \frac{1}{4} c_{2Y} (c_Y^2 Y_d^{\text{pr}} Y_d^{\text{pr}} (g_1^2 - 3 g_2^2) + s_Y^2 Y_u^{\text{pr}} Y_u^{\text{pr}} (g_1^2 + 3 g_2^2)) - \right. \right. \\ \left. \left. -\frac{1}{288} \sum_{\mathbf{p}} (g_1^4 (5 + 6 c_{2Y}^2) + 27 g_2^4 (5 - 2 c_{2Y}^2)) \right) LF_{4,-1}[m_q^{\mathbf{p}}] + \right. \\ \left. -\frac{1}{135} \sum_{\mathbf{p}} (g_1^4 + 27 g_2^4) LF_{5,-2}[m_q^{\mathbf{p}}] + \frac{1}{54} \sum_{\mathbf{p}} g_1^4 (4 + 9 c_{2Y}^2) LF_{3,0}[m_u^{\mathbf{p}}] - \right. \\ \left. -\frac{1}{36} \sum_{\mathbf{p}} g_1^4 (5 + 6 c_{2Y}^2) LF_{4,-1}[m_u^{\mathbf{p}}] + \frac{8}{135} \sum_{\mathbf{p}} g_1^4 LF_{5,-2}[m_u^{\mathbf{p}}] + \right. \\ c_{2Y} g_1^2 s_Y^2 Y_u^{\text{pr}} Y_u^{\text{pr}} LF_{3,0}[m_u^{\mathbf{r}}] - c_{2Y} g_1^2 s_Y^2 Y_u^{\text{pr}} Y_u^{\text{pr}} LF_{4,-1}[m_u^{\mathbf{r}}] + \\ \left. -\frac{1}{288} (g_1^4 (8 + 9 c_{4Y} (1 + c_{4Y}) - 9 s_{2Y}^4) + 3 g_2^4 (8 + 3 c_{4Y} (-3 + c_{4Y}) - 3 s_{2Y}^4) + \right. \\ \left. + 18 g_1^2 g_2^2 (c_{4Y} (-1 + c_{4Y}) - s_{2Y}^4)) LF_{3,0}[m_\oplus] + \frac{1}{96} (-g_1^4 (5 + 3 c_{4Y} (1 + c_{4Y}) - 3 s_{2Y}^4) + \right. \\ \left. + 3 g_2^4 (-5 - c_{4Y} (-3 + c_{4Y}) + s_{2Y}^4) + 6 g_1^2 g_2^2 (c_{4Y} - c_{4Y}^2 + s_{2Y}^4)) LF_{4,-1}[m_\oplus] + \right. \\ \left. -\frac{1}{45} (g_1^4 + 3 g_2^4) LF_{5,-2}[m_\oplus] + \frac{1}{36} (g_1^4 + 3 g_2^4) LF_{3,0}[\tilde{\mu}] + \frac{1}{24} (g_1^4 + 3 g_2^4) LF_{4,-1}[\tilde{\mu}] - \right. \\ \left. -\frac{1}{45} (g_1^4 + 3 g_2^4) LF_{5,-2}[\tilde{\mu}] + \frac{1}{8} g_1^4 (2 c_Y^4 + 3 s_Y^2 c_Y^2 + 2 s_Y^4) LF_{2,2,-1}[m_1, \tilde{\mu}] + \right. \\ \left. -\frac{1}{8} m_1 g_1^4 (m_1 (c_Y^4 + s_Y^4) + 4 s_Y \tilde{\mu} c_Y) LF_{2,2,0}[m_1, \tilde{\mu}] - \frac{1}{2} g_1^4 LF_{3,2,-2}[m_1, \tilde{\mu}] + \right. \\ \left. -\frac{3}{8} g_1^4 (-m_1^2 - 8 m_1 s_Y \tilde{\mu} c_Y - 4 s_Y^2 \tilde{\mu}^2 c_Y^2) LF_{3,2,-1}[m_1, \tilde{\mu}] - g_1^4 m_1^2 s_Y^2 \tilde{\mu}^2 c_Y^2 LF_{3,2,0}[m_1, \tilde{\mu}] + \right. \\ \left. + \frac{1}{4} g_1^4 LF_{3,3,-3}[m_1, \tilde{\mu}] + \frac{1}{4} g_1^4 (m_1^2 + 8 m_1 s_Y \tilde{\mu} c_Y + 4 s_Y^2 \tilde{\mu}^2 c_Y^2) LF_{3,3,-2}[m_1, \tilde{\mu}] + \right. \\ g_1^4 m_1^2 s_Y^2 \tilde{\mu}^2 c_Y^2 LF_{3,3,-1}[m_1, \tilde{\mu}] + \frac{1}{4} g_1^4 LF_{4,2,-3}[m_1, \tilde{\mu}] + \\ \left. + \frac{1}{4} g_1^4 (m_1^2 + 8 m_1 s_Y \tilde{\mu} c_Y + 4 s_Y^2 \tilde{\mu}^2 c_Y^2) LF_{4,2,-2}[m_1, \tilde{\mu}] + g_1^4 m_1^2 s_Y^2 \tilde{\mu}^2 c_Y^2 LF_{4,2,-1}[m_1, \tilde{\mu}] - \right. \\ g_2^4 LF_{2,1,0}[m_2, \tilde{\mu}] + \frac{1}{8} g_2^4 (4 c_Y^4 + 4 s_Y^2 (-2 + s_Y^2) + c_Y^2 (-8 + 19 s_Y^2)) LF_{2,2,-1}[m_2, \tilde{\mu}] + \\ \left. + \frac{1}{8} m_2 g_2^4 (m_2 (c_Y^4 - 8 s_Y^2 c_Y^2 + s_Y^4) + 4 s_Y \tilde{\mu} c_Y (-4 + 3 c_Y^2 + 3 s_Y^2)) LF_{2,2,0}[m_2, \tilde{\mu}] + \right. \\ \left. + \frac{1}{2} g_2^4 LF_{3,1,-1}[m_2, \tilde{\mu}] - \frac{1}{2} g_2^4 (c_Y^4 - s_Y^2 + s_Y^4 + c_Y^2 (-1 + 10 s_Y^2)) LF_{3,2,-2}[m_2, \tilde{\mu}] + \right. \\ \left. + \frac{1}{8} g_2^4 (-3 m_2^2 (5 c_Y^4 + 2 s_Y^2 c_Y^2 + 5 s_Y^4) - 8 m_2 s_Y \tilde{\mu} c_Y (-1 + 9 c_Y^2 + 9 s_Y^2) - 36 s_Y^2 \tilde{\mu}^2 c_Y^2) \right. \\ \left. LF_{3,2,-1}[m_2, \tilde{\mu}] - 3 g_2^4 m_2^2 s_Y^2 \tilde{\mu}^2 c_Y^2 LF_{3,2,0}[m_2, \tilde{\mu}] + \right. \\ \left. + \frac{1}{4} g_2^4 (c_Y^4 + 10 s_Y^2 c_Y^2 + s_Y^4) LF_{3,3,-3}[m_2, \tilde{\mu}] + \right. \\ \left. + \frac{1}{4} g_2^4 (m_2^2 (5 c_Y^4 + 2 s_Y^2 c_Y^2 + 5 s_Y^4) + 24 m_2 s_Y \tilde{\mu} c_Y + 12 s_Y^2 \tilde{\mu}^2 c_Y^2) LF_{3,3,-2}[m_2, \tilde{\mu}] + \right. \\ \left. + 3 g_2^4 m_2^2 s_Y^2 \tilde{\mu}^2 c_Y^2 LF_{3,3,-1}[m_2, \tilde{\mu}] + \frac{1}{4} g_2^4 (c_Y^4 + 10 s_Y^2 c_Y^2 + s_Y^4) LF_{4,2,-3}[m_2, \tilde{\mu}] + \right. \\ \left. + \frac{1}{4} g_2^4 (m_2^2 (5 c_Y^4 + 2 s_Y^2 c_Y^2 + 5 s_Y^4) + 24 m_2 s_Y \tilde{\mu} c_Y + 12 s_Y^2 \tilde{\mu}^2 c_Y^2) LF_{4,2,-2}[m_2, \tilde{\mu}] + \right. \\ \left. + 3 g_2^4 m_2^2 s_Y^2 \tilde{\mu}^2 c_Y^2 LF_{4,2,-1}[m_2, \tilde{\mu}] + \frac{3}{2} c_Y^4 Y_d^{\text{pr}} Y_d^{\text{st}} Y_d^{\text{pt}} Y_d^{\text{sr}} LF_{2,1,0}[m_d^{\mathbf{r}}, m_d^{\mathbf{t}}] - \right. \\ \left. -\frac{3}{2} c_Y^4 Y_d^{\text{pr}} Y_d^{\text{st}} Y_d^{\text{pt}} Y_d^{\text{sr}} LF_{3,1,-1}[m_d^{\mathbf{r}}, m_d^{\mathbf{t}}] + \right. \\ \left. -\frac{1}{6} g_1^2 (c_Y \bar{a}_d^{\text{pr}} - s_Y \tilde{\mu} \bar{Y}_d^{\text{pr}}) (c_Y a_d^{\text{pr}} - s_Y \tilde{\mu} Y_d^{\text{pr}}) LF_{2,2,0}[m_d^{\mathbf{r}}, m_q^{\mathbf{p}}] - \right. \\ \left. -\frac{1}{2} c_{2Y} g_1^2 (c_Y \bar{a}_d^{\text{pr}} - s_Y \tilde{\mu} \bar{Y}_d^{\text{pr}}) (c_Y a_d^{\text{pr}} - s_Y \tilde{\mu} Y_d^{\text{pr}}) LF_{3,1,0}[m_d^{\mathbf{r}}, m_q^{\mathbf{p}}] + \right. \\ \left. -\frac{1}{12} g_1^2 (-1 + 3 c_{2Y}) (c_Y \bar{a}_d^{\text{pr}} - s_Y \tilde{\mu} \bar{Y}_d^{\text{pr}}) (c_Y a_d^{\text{pr}} - s_Y \tilde{\mu$$